

### Tutorial 1

Prove the following statements using the definition of convexity.

For each proof, fix arbitrary  $u, v \in \mathbb{R}^d$  and  $\theta \in [0, 1]$ , and write  $w = \theta u + (1 - \theta)v$ . We need to prove  $F(w) \leq \theta F(u) + (1 - \theta)F(v)$ .

(a) If  $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$  are convex and  $\gamma_i \geq 0$ , then  $F(x) = \sum_{i=1}^r \gamma_i f_i(x)$  is convex.

**Solution:** (a) Convexity of each  $f_i$  gives  $f_i(w) \leq \theta f_i(u) + (1 - \theta)f_i(v)$ . Multiply by  $\gamma_i \geq 0$  and add:

$$\begin{aligned} F(w) &= \sum_{i=1}^r \gamma_i f_i(w) \\ &\leq \sum_{i=1}^r \gamma_i [\theta f_i(u) + (1 - \theta)f_i(v)] \\ &= \theta \sum_{i=1}^r \gamma_i f_i(u) + (1 - \theta) \sum_{i=1}^r \gamma_i f_i(v) = \theta F(u) + (1 - \theta)F(v). \end{aligned}$$

This is the required inequality, so  $F$  is convex. Nonnegative weights preserve the inequality direction; they do *not* need to sum to 1.

(b) If  $g : \mathbb{R}^m \rightarrow \mathbb{R}$  is convex,  $M \in \mathbb{R}^{m \times d}$ , and  $q \in \mathbb{R}^m$ , then  $F(x) = g(Mx + q)$  is convex.

**Solution:** (b) Since  $\theta q + (1 - \theta)q = q$ ,

$$\begin{aligned} Mw + q &= \theta Mu + (1 - \theta)Mv + q \\ &= \theta(Mu + q) + (1 - \theta)(Mv + q). \end{aligned}$$

Apply convexity of  $g$  to the two points  $Mu + q$  and  $Mv + q$ :

$$\begin{aligned} F(w) &= g(\theta(Mu + q) + (1 - \theta)(Mv + q)) \\ &\leq \theta g(Mu + q) + (1 - \theta)g(Mv + q) = \theta F(u) + (1 - \theta)F(v). \end{aligned}$$

Thus  $F$  is convex. No sign restriction on the entries of  $M$  is needed.

(c) If  $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$  are convex, then  $F(x) = \max_{1 \leq i \leq r} f_i(x)$  is convex.

**Solution:** (c) Fix any index  $i$ . First apply convexity of  $f_i$ :

$$f_i(w) \leq \theta f_i(u) + (1 - \theta) f_i(v).$$

By definition,  $f_i(u) \leq F(u)$  and  $f_i(v) \leq F(v)$ . Since  $\theta$  and  $1 - \theta$  are nonnegative,

$$\theta f_i(u) + (1 - \theta) f_i(v) \leq \theta F(u) + (1 - \theta) F(v).$$

Combining the two inequalities, *every*  $f_i(w)$  is at most  $\theta F(u) + (1 - \theta) F(v)$ . This common bound is independent of  $i$ , so taking the maximum gives

$$F(w) = \max_{1 \leq i \leq r} f_i(w) \leq \theta F(u) + (1 - \theta) F(v).$$

Thus  $F$  is convex. The maximizing index need not be the same at  $u$ ,  $v$ , and  $w$ .

1. Determine whether each statement is *True* or *False*. For a true statement, give a short justification. For a false statement, give a counterexample and verify it.
  - (a) If  $g : \mathbb{R} \rightarrow \mathbb{R}$  is convex, then  $h(t) = g(1 - 2t)$  is convex.
  - (b) If  $f_1, f_2 : \mathbb{R} \rightarrow \mathbb{R}$  are convex, then  $h(t) = \min\{f_1(t), f_2(t)\}$  is convex.
  - (c) A symmetric matrix whose entries are all strictly positive is positive definite.
  - (d) If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is twice continuously differentiable and  $t^*$  is a strict local minimizer, then  $f''(t^*) > 0$ .
  - (e) For differentiable  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  and  $\nabla f(x) \neq 0$ ,  $p = -3\nabla f(x)$  is a descent direction at  $x$ .
  - (f) If  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is  $L$ -smooth with  $L > 0$  and  $\nabla f(x) \neq 0$ , then

$$f\left(x - \frac{1}{L} \nabla f(x)\right) < f(x).$$

**Solution: (a) True.** Apply the affine-composition rule with  $M = -2$  and  $q = 1$ . The input map  $t \mapsto 1 - 2t$  is affine; a negative slope is allowed.

**(b) False.** Let  $f_1(t) = t$  and  $f_2(t) = -t$ . Both are affine, so their convexity inequalities hold with equality. Their minimum is  $h(t) = -|t|$ . At  $u = -1$ ,  $v = 1$ , and  $\theta = 1/2$ ,

$$h(\tfrac{1}{2}u + \tfrac{1}{2}v) = h(0) = 0, \quad \tfrac{1}{2}h(-1) + \tfrac{1}{2}h(1) = \tfrac{1}{2}(-1) + \tfrac{1}{2}(-1) = -1.$$

Convexity would require  $0 \leq -1$ , which is false.

**(c) False.** Take

$$Q = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad z = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \neq 0.$$

The matrix is symmetric and every entry is positive, but

$$Qz = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad z^T Qz = (1, -1) \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -2 < 0.$$

Positive definiteness requires  $z^T Qz > 0$  for *every* nonzero  $z$ , so this one direction disproves it.

**(d) False.** Let  $f(t) = t^6$ . For every  $t \neq 0$ ,  $f(t) > f(0) = 0$ , so 0 is a strict local (indeed global) minimizer. However,

$$f'(t) = 6t^5, \quad f''(t) = 30t^4, \quad f''(0) = 0.$$

At a stationary point, positive second derivative is sufficient for a strict local minimum, but it is not necessary.

**(e) True.** The directional derivative along  $p$  is

$$\nabla f(x)^T p = \nabla f(x)^T [-3\nabla f(x)] = -3 \|\nabla f(x)\|^2 < 0.$$

The last inequality uses  $\nabla f(x) \neq 0$ . Hence  $p$  is a descent direction: sufficiently small positive steps along  $p$  decrease  $f$ .

**(f) True.** Apply the descent lemma with  $u = x$  and  $v = x - \nabla f(x)/L$ :

$$\begin{aligned} f(v) &\leq f(x) + \nabla f(x)^T (v - x) + \frac{L}{2} \|v - x\|^2 \\ &= f(x) - \frac{1}{L} \|\nabla f(x)\|^2 + \frac{L}{2} \frac{1}{L^2} \|\nabla f(x)\|^2 \\ &= f(x) - \frac{1}{2L} \|\nabla f(x)\|^2 < f(x). \end{aligned}$$

The strict inequality follows from  $L > 0$  and  $\nabla f(x) \neq 0$ . Convexity is not required.

2. For a linear prediction model  $a_i^T x$ , let

$$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, y_1 = 1, \quad a_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, y_2 = 2, \quad a_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, y_3 = 1.$$

We minimize the mean squared-error objective

$$F(x) = \frac{1}{2n} \sum_{i=1}^n (a_i^T x - y_i)^2, \quad n = 3.$$

- (a) Write  $F$  explicitly in terms of  $x_1, x_2$ , and as  $F(x) = \frac{1}{6} \|Ax - y\|^2$ . State the dimensions of  $A$  and  $y$ .
- (b) Use the chain rule to find  $\nabla F(x)$  in matrix form, then write its two components explicitly.
- (c) At  $x^{(0)} = (0, 0)^T$ , calculate the directional derivatives along

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Which is a descent direction? Find the steepest descent direction among all unit vectors.

**Solution:** (a) The predictions are  $x_1$ ,  $x_1 + x_2$ , and  $x_2$ . Thus

$$F(x) = \frac{1}{6} [(x_1 - 1)^2 + (x_1 + x_2 - 2)^2 + (x_2 - 1)^2].$$

Stacking the feature vectors as *rows* gives

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 2}, \quad y = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \in \mathbb{R}^3, \quad Ax - y = \begin{pmatrix} x_1 - 1 \\ x_1 + x_2 - 2 \\ x_2 - 1 \end{pmatrix}.$$

The sum of the three squared residuals is  $\|Ax - y\|^2$ , hence  $F(x) = \frac{1}{6} \|Ax - y\|^2$ .

(b) Write  $r(x) = Ax - y$  and  $\phi(r) = \frac{1}{6} \|r\|^2$ . The Jacobian of  $r$  is  $A$ . Since  $\partial\phi/\partial r_i = 2r_i/6 = r_i/3$ , we have  $\nabla\phi(r) = r/3$ . The chain rule gives

$$\begin{aligned} \nabla F(x) &= A^T \nabla\phi(r(x)) = \frac{1}{3} A^T (Ax - y) \\ &= \frac{1}{3} \begin{pmatrix} (x_1 - 1) + (x_1 + x_2 - 2) \\ (x_1 + x_2 - 2) + (x_2 - 1) \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2x_1 + x_2 - 3 \\ x_1 + 2x_2 - 3 \end{pmatrix}. \end{aligned}$$

The gradient has two components, one for each model parameter.

**Solution: (c)** At the starting point,  $\nabla F(x^{(0)}) = \frac{1}{3}(-3, -3)^T = (-1, -1)^T$ . The directional derivatives are

$$\begin{aligned}\nabla F(x^{(0)})^T v_1 &= \frac{-1-1}{\sqrt{2}} = -\sqrt{2} < 0, \\ \nabla F(x^{(0)})^T v_2 &= \frac{-1+1}{\sqrt{2}} = 0.\end{aligned}$$

Thus  $v_1$  is a descent direction. The derivative along  $v_2$  gives no first-order decrease; in fact, direct substitution yields  $F(tv_2) = 1 + t^2/6 > F(0) = 1$  for  $t \neq 0$ , so  $v_2$  is not a descent direction.

For any unit vector  $v$ , Cauchy–Schwarz gives

$$\nabla F(x^{(0)})^T v \geq -\|\nabla F(x^{(0)})\| \|v\| = -\sqrt{2}.$$

Equality is attained at

$$v = -\frac{\nabla F(x^{(0)})}{\|\nabla F(x^{(0)})\|} = \frac{1}{\sqrt{2}}(1, 1)^T = v_1.$$

Therefore  $v_1$  is the steepest descent direction among all unit vectors.

3. Consider

$$f(x_1, x_2) = x_1^2 - 2x_1x_2 + \frac{2}{3}x_2^3.$$

- (a) Compute  $\nabla f(x)$  and  $\nabla^2 f(x)$ , and find all stationary points.
- (b) Classify each stationary point as a local minimizer, a local maximizer, or neither. You may test a Hessian using  $z^T \nabla^2 f(x) z$ ; no eigenvalue calculation is required.
- (c) Does  $f$  have a global minimizer on  $\mathbb{R}^2$ ? Justify your answer by examining  $f(0, -t)$  as  $t \rightarrow +\infty$ .

**Solution: (a)** Differentiate with respect to each variable:

$$\nabla f(x) = \begin{pmatrix} 2x_1 - 2x_2 \\ -2x_1 + 2x_2^2 \end{pmatrix}, \quad \nabla^2 f(x) = \begin{pmatrix} 2 & -2 \\ -2 & 4x_2 \end{pmatrix}.$$

Both gradient components must vanish. They give  $x_1 = x_2$  and  $x_1 = x_2^2$ , so

$$x_2 = x_2^2 \iff x_2(x_2 - 1) = 0 \iff x_2 \in \{0, 1\}.$$

Using  $x_1 = x_2$ , the stationary points are  $(0, 0)^T$  and  $(1, 1)^T$ .

**Solution: (b)** At the origin,  $H_0 = \begin{pmatrix} 2 & -2 \\ -2 & 0 \end{pmatrix}$ , with quadratic form  $z^T H_0 z = 2z_1^2 - 4z_1 z_2$ . For  $z = (1, 0)^T$  the value is  $2 > 0$ , while for  $z = (1, 1)^T$  it is  $2 - 4 = -2 < 0$ . Hence the Hessian is indefinite. We can verify the classification directly: for  $0 < |t| < 1$ ,

$$f(t, 0) = t^2 > 0 = f(0, 0), \quad f(t, t) = -t^2 + \frac{2}{3}t^3 = t^2(-1 + \frac{2}{3}t) < 0.$$

Both paths approach the origin as  $t \rightarrow 0$ . There are larger and smaller values arbitrarily nearby, so the origin is a *saddle point*, neither a local minimum nor a local maximum.

At  $(1, 1)^T$ , the Hessian is  $H_1 = \begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix}$ . Complete the square in its quadratic form:

$$z^T H_1 z = 2z_1^2 - 4z_1 z_2 + 4z_2^2 = 2(z_1 - z_2)^2 + 2z_2^2.$$

This is nonnegative, and it equals zero only if  $z_2 = 0$  and  $z_1 - z_2 = 0$ , that is,  $z = 0$ . Therefore it is positive for every  $z \neq 0$ , proving  $H_1 \succ 0$ . Together with  $\nabla f(1, 1) = 0$ , the second-order sufficient condition gives a strict local minimizer, with value  $f(1, 1) = 1 - 2 + \frac{2}{3} = -\frac{1}{3}$ .

(c) Along the specified path, for  $t > 0$ ,  $f(0, -t) = 0 - 0 + \frac{2}{3}(-t)^3 = -\frac{2}{3}t^3 \rightarrow -\infty$  as  $t \rightarrow +\infty$ . Thus  $f$  is unbounded below and has no global minimizer. In particular, the local minimum in part (b) is not a global minimum.

4. Let

$$f(x) = \frac{1}{2}(4x_1^2 + x_2^2), \quad x^{(0)} = (1, 2)^T, \quad x^{(1)} = x^{(0)} - \alpha \nabla f(x^{(0)}).$$

Recall that  $f$  is  $L$ -smooth if  $\|\nabla f(u) - \nabla f(v)\| \leq L \|u - v\|$  for all  $u, v$ .

- (a) Find  $\nabla f(x)$  and show *directly from this definition* that the smallest possible smoothness constant is  $L = 4$ .
- (b) Use  $\alpha = 1/L$  to calculate  $x^{(1)}$  and  $f(x^{(1)})$ . Compare the actual value with the descent-lemma upper bound

$$f(x^{(0)}) - \frac{1}{2L} \|\nabla f(x^{(0)})\|^2.$$

- (c) Restart from the same  $x^{(0)}$  and instead use  $\alpha = 3/4$ . Calculate the new point and objective value. Explain why this does not contradict the fact that  $-\nabla f(x^{(0)})$  is a descent direction.

**Solution: (a)** We have  $\nabla f(x) = (4x_1, x_2)^T$ . For any  $u, v \in \mathbb{R}^2$ , write  $d = u - v$ . Then

$$\begin{aligned}\nabla f(u) - \nabla f(v) &= (4d_1, d_2)^T, \\ \|\nabla f(u) - \nabla f(v)\|^2 &= 16d_1^2 + d_2^2 \\ &\leq 16(d_1^2 + d_2^2) = 16\|u - v\|^2.\end{aligned}$$

Taking square roots gives the Lipschitz inequality with  $L = 4$ . For  $u - v = (1, 0)^T$ , however,

$$\frac{\|\nabla f(u) - \nabla f(v)\|}{\|u - v\|} = \frac{\|(4, 0)^T\|}{\|(1, 0)^T\|} = 4.$$

Any valid  $L$  must therefore be at least 4. The smallest possible constant is exactly  $L = 4$ .

**(b)** First calculate

$$\nabla f(x^{(0)}) = (4, 2)^T, \quad f(x^{(0)}) = \frac{1}{2}(4 + 4) = 4, \quad \|\nabla f(x^{(0)})\|^2 = 4^2 + 2^2 = 20.$$

Using  $\alpha = 1/4$ ,

$$x^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{1}{4} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 - 1 \\ 2 - 1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ 3/2 \end{pmatrix}.$$

Consequently,

$$f(x^{(1)}) = \frac{1}{2}[4(0)^2 + (3/2)^2] = \frac{9}{8}.$$

The descent-lemma upper bound is

$$4 - \frac{1}{2(4)}(20) = 4 - \frac{5}{2} = \frac{3}{2}.$$

Indeed,  $9/8 \leq 3/2 < 4$ . The bound guarantees a decrease; it need not equal the actual objective value.

**(c)** Restarting from  $x^{(0)}$ , the larger step gives

$$\tilde{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{3}{4} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 - 3 \\ 2 - 3/2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1/2 \end{pmatrix},$$

and

$$f(\tilde{x}^{(1)}) = \frac{1}{2}[4(-2)^2 + (1/2)^2] = \frac{1}{2}(16 + \frac{1}{4}) = \frac{65}{8} > 4.$$

The direction is still a descent direction because  $\nabla f(x^{(0)})^T[-\nabla f(x^{(0)})] = -20 < 0$ . But this only guarantees a decrease for sufficiently small positive steps. The descent-lemma guarantee applies when  $0 < \alpha < 2/L = 1/2$ ; the trial step  $\alpha = 3/4$  lies outside that range and overshoots.